SEAT No.________

[5, 6, A-6]

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Sardar Patel University

B.Sc. (CA & IT) External Examination (CBCS) &

M. Sc.(Integrated) IInd Semester (Information Technology)

PS02EIIT01 – Digital Electronics

22nd April, Saturday - 2017

Time : 10:00 A.M. to 12:00 P.M. Total Marks :70

Q.1 Select an appropriate option. **10**

1. A combinational circuit that performs the arithmetic addition of two bits is called ________.
   (a) Full Adder (b) Half Adder (c) Binary Adder (d) Decoder
2. In half adder XOR gate's output is __________.
   (a) carry (b) sum (c) remainder (d) none
3. A _____ is a combinational circuit that converts binary information from the 2^n coded inputs to an output.
   (a) carry (b) sum (c) remainder (d) none
4. Which is used as data selector?
   (a) Encoder (b) Decoder (c) Multiplexer (d) Flip-Flop
5. A 16 X 1 line multiplexer requires _____ data select line.
   (a) 3 (b) 4 (c) 8 (d) 16
6. The 16 bit register is separated into groups of 4 bit where each groups is called ________.
   (a) BCD (b) Nibble (c) Half byte (d) None
7. In k-map, pair eliminates _______ variable.
   (a) one (b) two (c) three (d) four
8. Don't care conditions are marked as ____ in the output column of the function table.
   (a) 1 (b) 0 (c) X (d) none of these
9. In D flip-flop, when CLK is low then input is _______.
   (a) High (b) Low (c) Don't care (d) Not change
10. In shift right register, the arrival of the first rising clock edge sets the ______ flip-flop.
    (a) left (b) right (c) up (d) down

Q.2 Answer the following questions. (Attempt any TEN) **20**

1. Describe half adder in short.
2. Draw the circuit of decoder.
3. Describe binary adder in short.
4. What is multiplexer?
5. Explain comparator in brief.
6. Draw the circuit of Seven Segment Decoder.
7. Explain Sum of Product in detail.

8. Explain K-Map for 2 variable with example.
9. Simplify using k-map F(A,B,C)=E(1,2,5).
10. Define flip-flop.
11. Draw circuit diagram of D flip-flop.
12. Explain shift left register in brief.

Q.3	[a] Explain de Morgan's first and second theorem.	**6**
	[b] Explain full-adder with circuit and truth able.	**4**
	OR	
Q.3	[a] Explain binary adder-subtracter in detail.	**6**
	[b] Write notes on encoder.	**4**
Q.4	[a] Explain 8x1 line multiplexer with circuit in detail.	**5**
	[b] Explain 4x1 line de-multiplexer with circuit in detail.	**5**
	OR	
Q.4	[a] Write a short note on Nibble Multiplexer with circuit.	**5**
	[b] Explain 8x1 line de-multiplexer with circuit in detail.	**5**
Q.5	What is k-map? Explain octet with example. Simplify this using k-map F(A,B,C,D)=Σ(1,3,5,6,8,11,15).	**10**
	OR	
Q.5	Explain pair and quad of k-map in detail. Simplify this using k-map F(A,B,C,D)= Σ (1,2,5,6,8,12,14).	**10**
Q.6	[a] Explain JK Master Slave flip-flop in detail.	**5**
	[b] Explain ring counters in detail.	**5**
	OR	
Q.6	[a] Explain controlled buffer register.	**5**
	[b] Explain shift right circuit in detail.	**5**

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